

15 Ans: D

Key thing to remember is, period of oscillation = 2 × period of energy-time graph

Therefore, period of oscillation = 400×10^{-3} s

Hence, $\omega = 2\pi / T = 15.708 \text{ rad s}^{-1}$

Also, $\max U = 1.0$

$$\frac{1}{2} m x_0^2 \omega^2 = 1.0$$

$$\frac{1}{2} (0.0900)(x_0^2)(15.708^2) = 1.0$$

$$x_0 = 0.300$$

Hence, $\max a = x_0 \omega^2 = 74 \text{ m s}^{-2}$

